Lab Manual - LAB 11

Semester V – Fall

Group A and B

Course Title: Computer Networks

Bachelor of Science in Software Engineering

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LAB 11: Connecting Two LANs Through a Router

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Connecting Two LANs Through a Router

In this lab, we will design a small university network consisting of two separate buildings. Each building will have its own Local Area Network (LAN), computers, and switch. A router will connect the two LANs so that computers in one building can communicate with computers in the other.

We will use both the **Physical Workspace** and **Logical Workspace** in Cisco Packet Tracer.

The Physical Workspace will be used to represent the actual buildings and the locations of networking devices. The Logical Workspace will be used to design the network topology, connect devices, configure IP addresses, and test communication.

Learning Objectives

By the end of this lab, students will be able to:

- Create two separate LANs in Cisco Packet Tracer.
- Use the Physical Workspace to represent two buildings and organize their networking devices.
- Connect two LANs through a router using separate router interfaces.
- Configure router interface IP addresses and PC default gateways.
- Test inter-network communication using ping.
- Identify and troubleshoot common connectivity problems.

Basic Concepts

What is a LAN?

A **Local Area Network (LAN)** connects devices within a limited geographical area, such as a computer lab, office, department, or building. For example, computers in the Computer Science Department connected through a switch form a LAN.

Why do we need a router?

A router connects different IP networks and forwards packets between them.

For example, a computer in the Computer Science Department may need to communicate with a computer in the Administration Department. If these computers belong to different IP networks, a router can forward traffic between them.

What is a default gateway?

A **default gateway** is the IP address of the router interface that a computer uses to send traffic to destinations outside its local network. For this lab:

- Computers in Building A will use 192.168.10.1 as their default gateway.
- Computers in Building B will use 192.168.20.1 as their default gateway.

Physical Workspace vs. Logical Workspace

Physical Workspace	Logical Workspace
Represents the physical locations of devices.	Represents the network topology and connections.
Used to organize buildings, wiring closets, and equipment locations.	Used to configure IP addresses, interfaces, and network settings.
Helps visualize where network devices are installed.	Helps visualize how devices communicate.
Example: A switch installed in Building A's wiring closet.	Example: That switch connected to a router interface.

Important: Moving a device into a building does not automatically assign its IP address or configure its network. Physical placement and logical configuration serve different purposes.

Scenario: A University with Two Buildings

A university has two buildings:

- **Building A:** Computer Science Department
- **Building B:** Administration Department

Each building contains two computers and one switch. A router connects the two networks. The network administrator must ensure that computers can communicate within their own building and across buildings.

Required Network Design

Component	Building A	Building B
Department	Computer Science	Administration
Computers	PC-A1, PC-A2	PC-B1, PC-B2
Switch	Switch A	Switch B
Network address	192.168.10.0/24	192.168.20.0/24
Router gateway	192.168.10.1	192.168.20.1

The router will use one Ethernet interface for each LAN.

Connection Plan

- PC-A1 → Switch A
- PC-A2 → Switch A
- Switch A → Router interface G0/0
- Router interface G0/1 → Switch B
- PC-B1 → Switch B
- PC-B2 → Switch B

The router connects the two different IP networks.

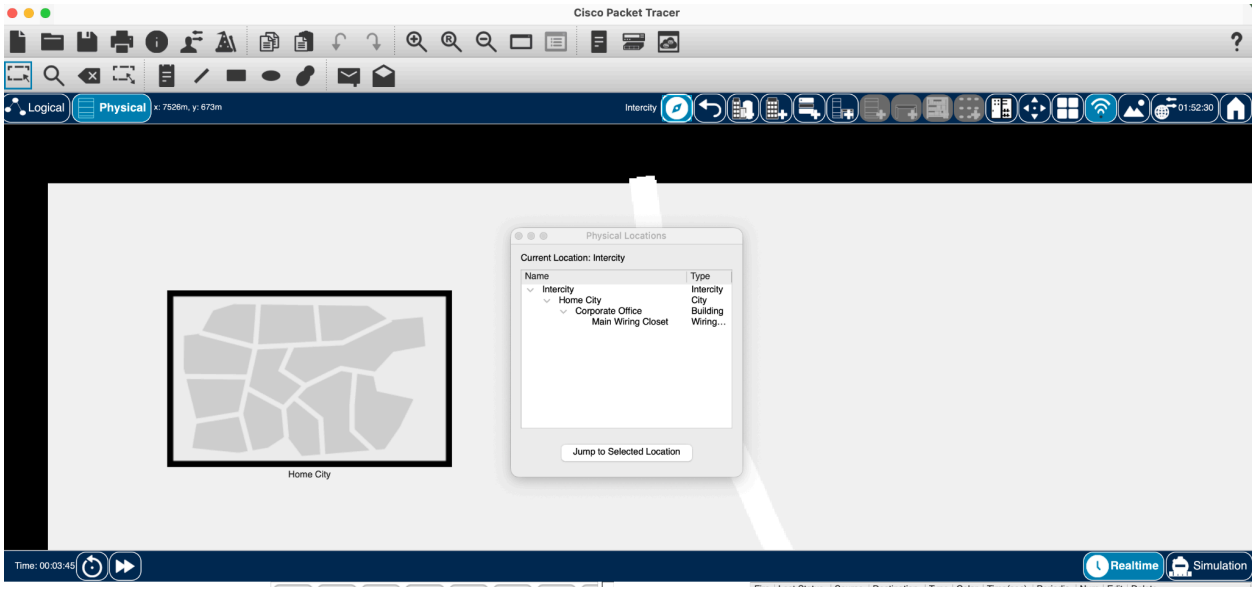
Physical Workspace and Network Construction

Part A: Explore the Physical Workspace

Step 1: Open Physical Workspace

1. Open Cisco Packet Tracer.
2. Open a new or existing project.
3. Locate the **Physical** and **Logical** workspace options near the top of the application.
4. Click **Physical**.
5. Explore the available physical locations.

Depending on the Packet Tracer version, you may see locations such as Intercity, City, Building, and Wiring Closet.

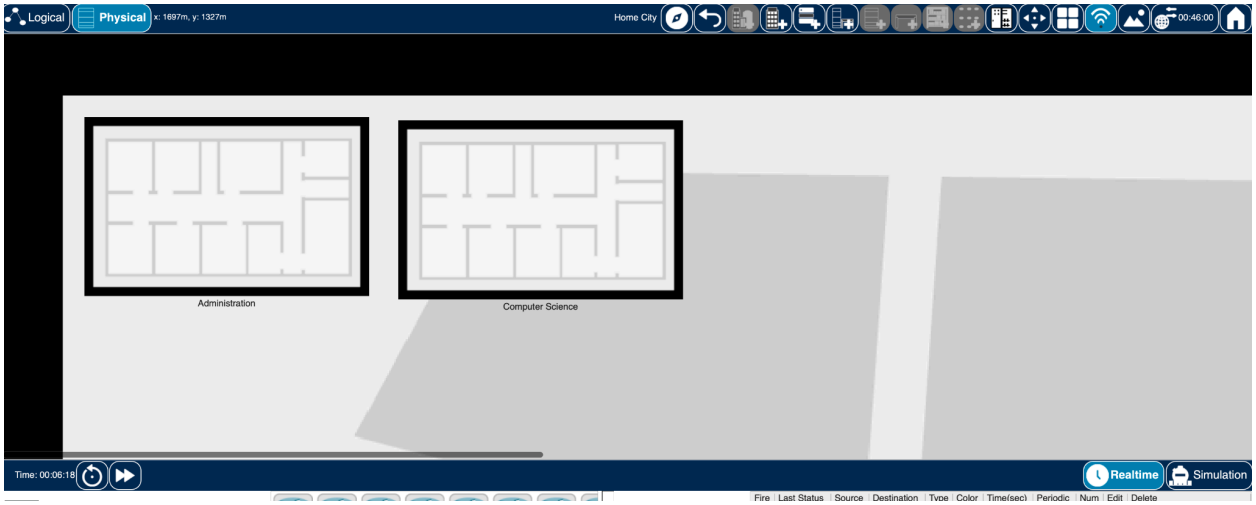


Step 2: Open the City

1. Navigate to the city-level view, such as Home City, if available.
2. Examine the buildings shown in the workspace.
3. Identify the options for adding or renaming buildings, if provided by your version.
4. Create or select two buildings for the activity.

Use these building names:

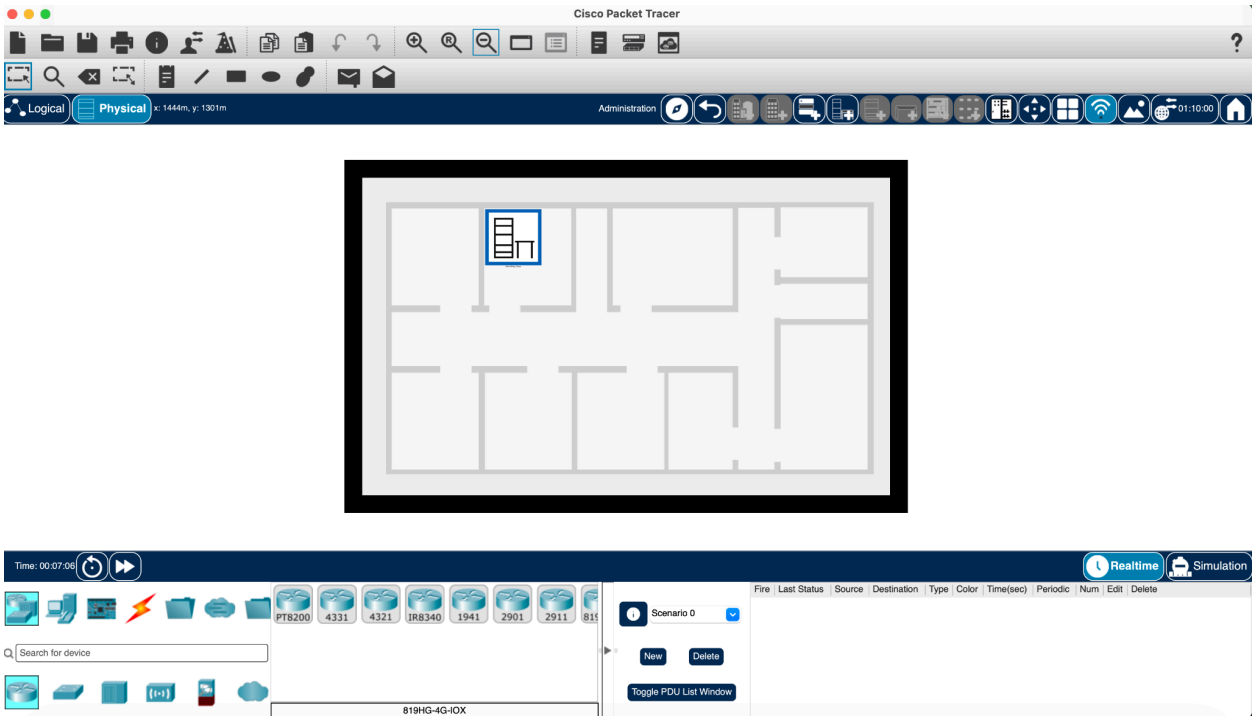
- Building A - Computer Science
- Building B - Administration



Step 3: Open Each Building

1. Open Building A.
2. Explore its internal physical layout.
3. Locate or create a wiring closet if the option is available.
4. Repeat the process for Building B.

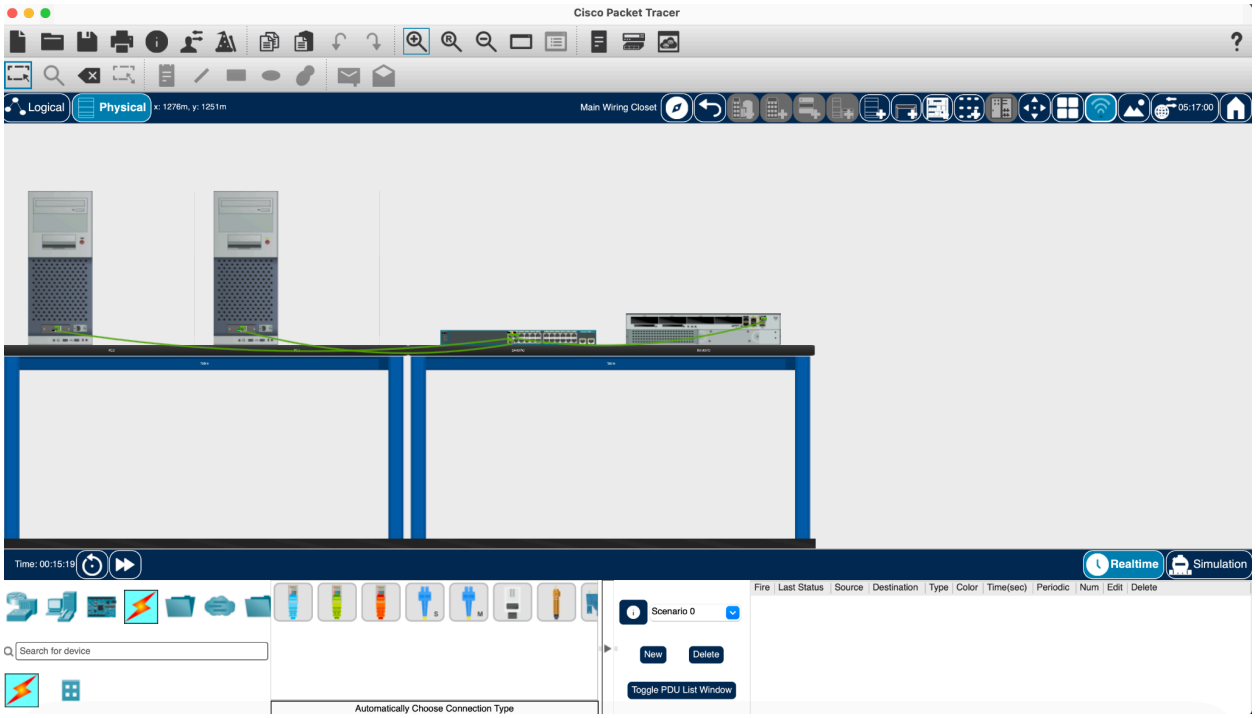
A **wiring closet** is a dedicated location where network equipment, such as switches and sometimes routers, is installed. In a real university, the wiring closet may contain equipment that distributes network connectivity throughout the building.



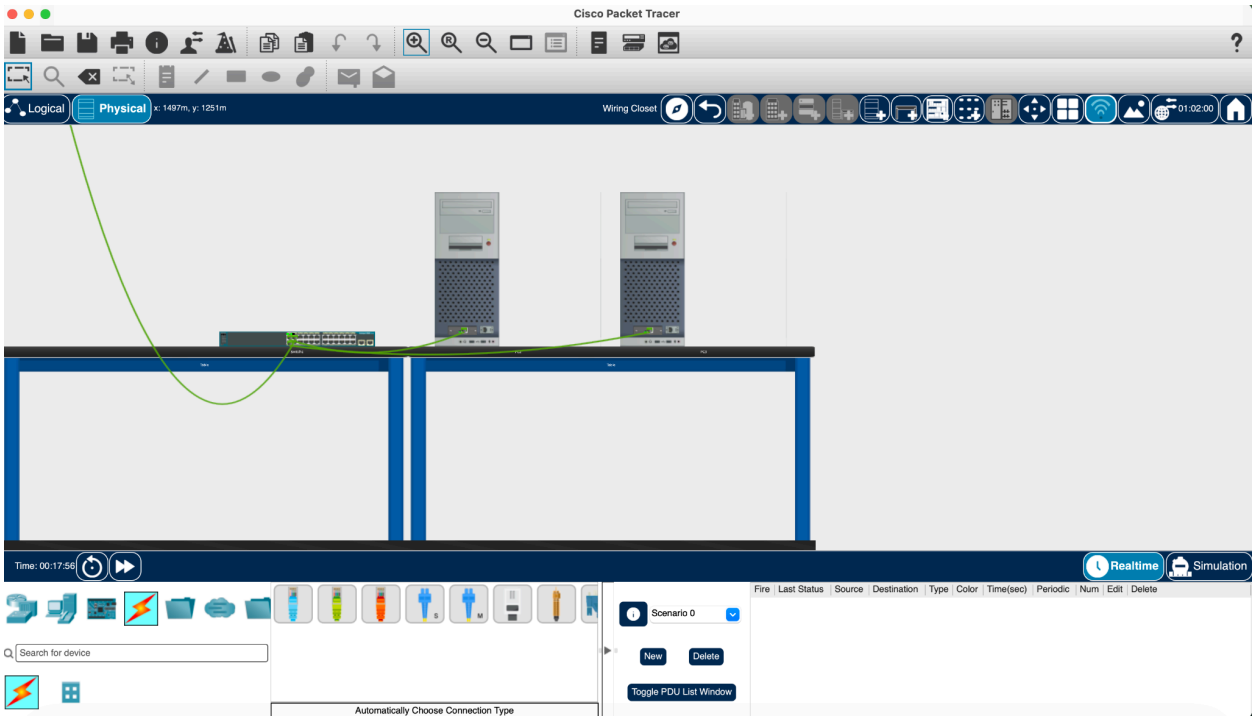
Step 4: Understand Device Placement

Devices should be organized according to their physical location.

- Building A should contain PC-A1, PC-A2, and Switch A.
- Building B should contain PC-B1, PC-B2, and Switch B.
- The router may be placed in Building A's wiring closet for this exercise.

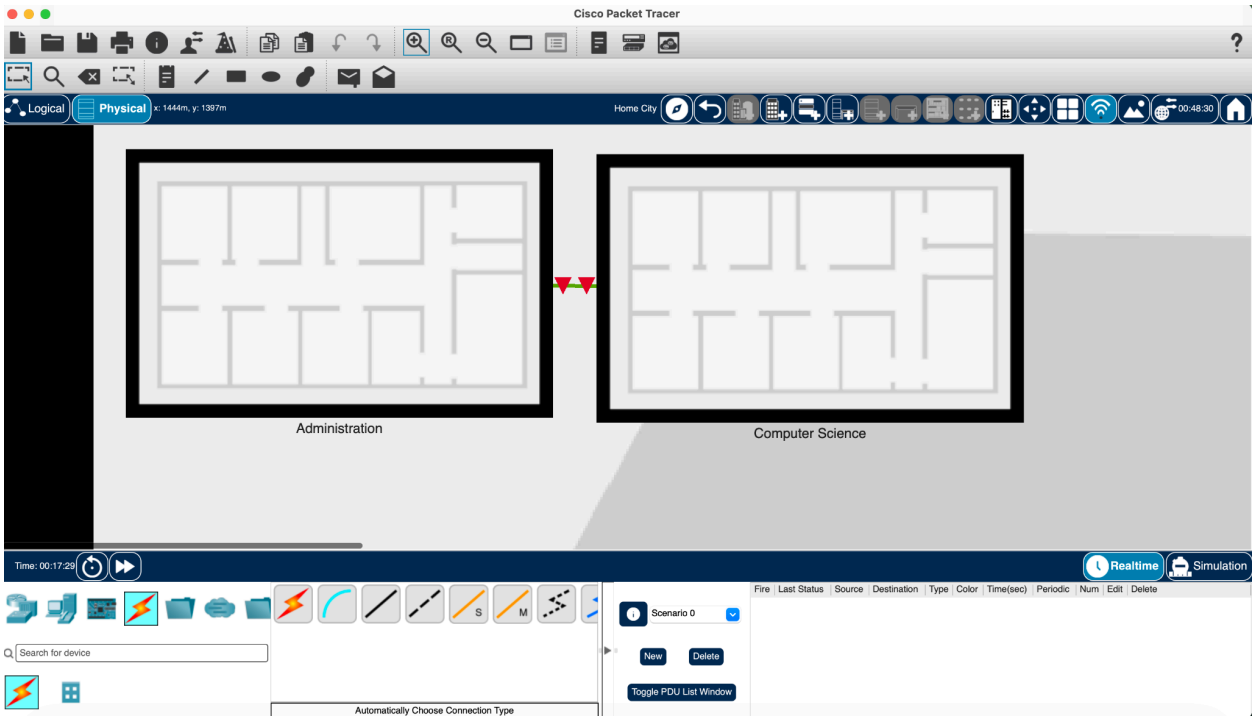


The router's second interface will connect to Switch B in Building B.



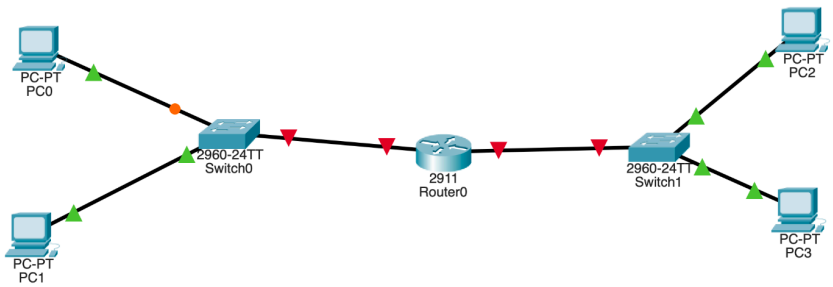
This arrangement represents a network in which the router is physically located in one building but connects two separate LANs.

Note: The exact controls for moving or placing devices depend on the Packet Tracer version. If you build the network in Logical Workspace first, use the Physical Workspace navigation and device-placement controls available in your version to organize the same devices into their respective locations.



Part B: Build the Logical Network

Now switch to **Logical Workspace**.



Part C: Configure the PCs

For each PC:

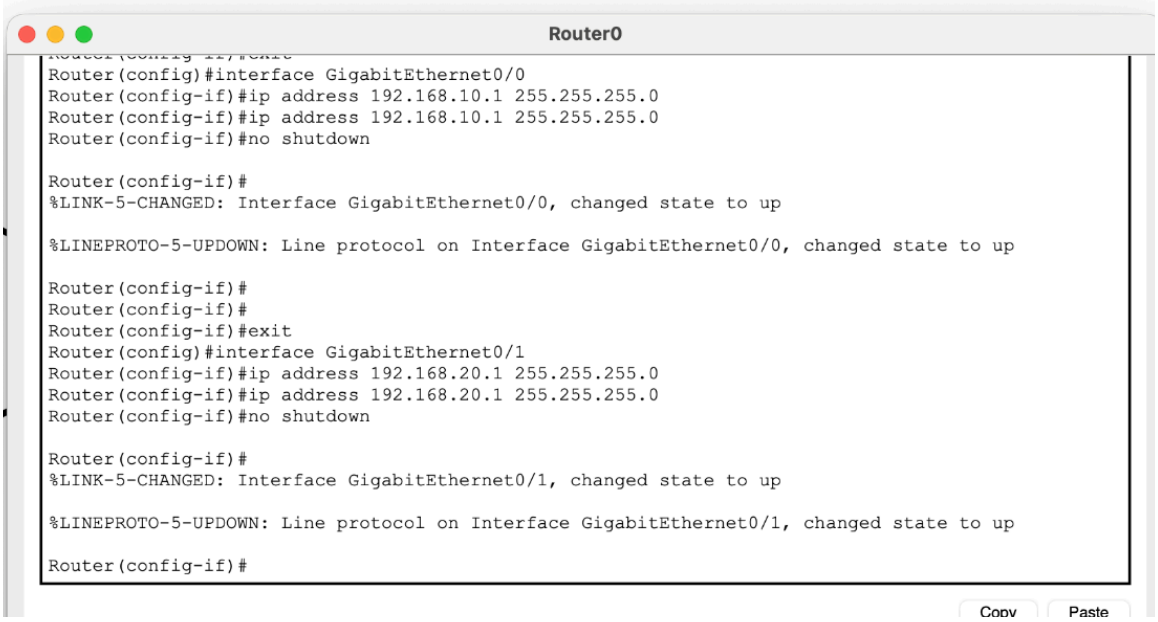
1. Click the PC.
2. Open **Desktop**.
3. Select **IP Configuration**.
4. Enter the IP address, subnet mask, and default gateway shown below.

IP Addressing Table

Device	IP Address	Subnet Mask	Default Gateway
PC-A1	192.168.10.2	255.255.255.0	192.168.10.1
PC-A2	192.168.10.3	255.255.255.0	192.168.10.1
PC-B1	192.168.20.2	255.255.255.0	192.168.20.1
PC-B2	192.168.20.3	255.255.255.0	192.168.20.1

Part D: Configure the Router

The router needs an IP address on each of its two connected LAN interfaces.



Important: Each router interface must have an address in the network connected to that interface. Do not assign the same IP address to both interfaces.

Part E: Test Network Connectivity

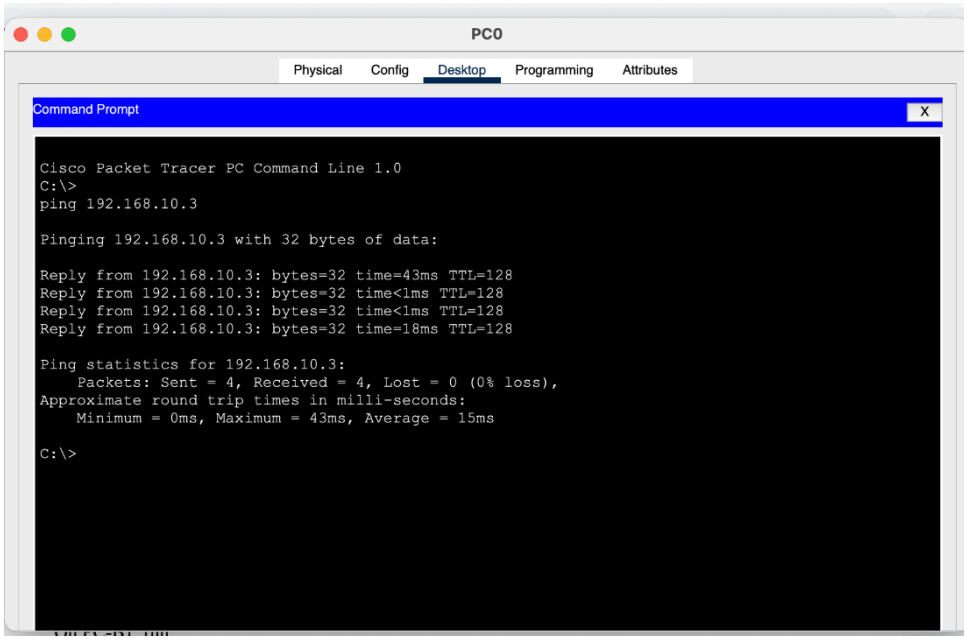
After configuring the PCs and router, perform the following tests.

Test 1: Check Communication Within Building A

On PC-A1:

1. Open **Desktop → Command Prompt**.
2. Type: ping 192.168.10.3

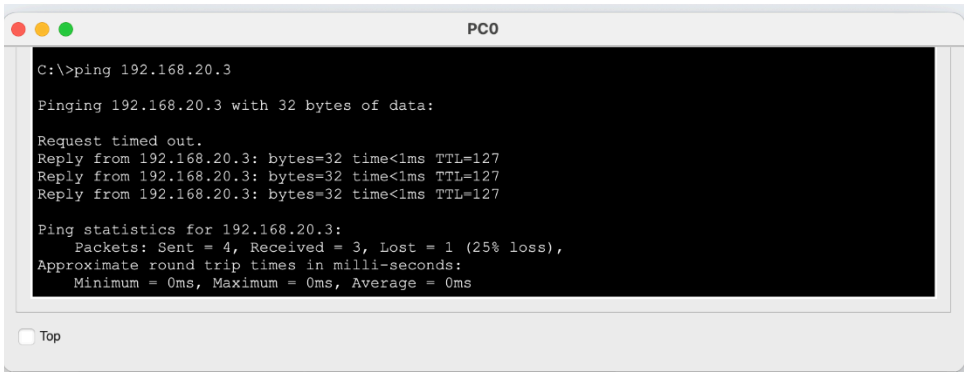
This tests communication from PC-A1 to PC-A2 within the same LAN.



Test 2: Check Communication Within Building B

On PC-B1, run: ping 192.168.20.3

This tests communication between PC-B1 and PC-B2 within Building B.



Conclusion

In this lab, we created two separate LANs and connect them through a router. We learned how to represent network equipment in Physical Workspace, construct a logical topology, configure router interfaces, assign PC default gateways, and test connectivity.

The most important concept is that **a router enables communication between different IP networks**. The physical workspace represents where equipment is located, while the logical workspace represents how devices are connected and configured.
